Isaac Malsky

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imalsky
Google Scholar

Appointments

2026–present **Pipeline Scientist**, *SETI Institute*, Mountain View, California
TOLIMAN mission data pipeline and exoplanet atmospheric modeling

2024–2026 **Postdoctoral Researcher**, *NASA Jet Propulsion Laboratory*, Pasadena, California
NASA Postdoctoral Program Fellow, Science Division

2019–2024 **Graduate Researcher**, *University of Michigan*, Ann Arbor, Michigan
Rauscher Lab

2015–2019 **Research Assistant**, *University of Chicago*, Chicago, Illinois
Rogers Lab

2018–2019 **Post-baccalaureate Researcher**, *Argonne National Laboratory*, Lemont, Illinois
Chan Lab, Center for Nanoscale Materials

Education

2019–2024 **Ph.D., Astronomy and Astrophysics**, *University of Michigan*, Ann Arbor, Michigan
Advisor: Prof. Emily Rauscher.
Dissertation: *Modeling the 3D Atmospheric Structure of Hot Gaseous Planets*.

2014–2018 **B.A., Physics, with Honors**, *University of Chicago*, Chicago, Illinois
Thesis: *The Coupled Thermal and Compositional Evolution of Planet Envelopes*.

Publications

6 published first-author papers, 1 in review, 958 total citations, and an ADS h-index of 17.

First Author

Malsky, I., Zhang, X., Kataria, T., et al. Accelerating Chemical Kinetics for Exoplanet Atmospheres Using Neural Networks. (2026). *ApJ*.

Malsky, I., Kataria, T., Batalha, N. E., and Graham, M. Accelerating Radiative Transfer for Planetary Atmospheres by Orders of Magnitude with a Transformer-Based Machine Learning Model. arXiv:2510.27050. In review, *ApJ*.

Malsky, I., Rauscher, E., Stevenson, K., et al. Clouds and Hazes in GJ 1214 b's Metal-rich Atmosphere. (2025). *AJ*.

Malsky, I., Rauscher, E., Roman, M. T., et al. A Direct Comparison between the Use of Double Gray and Multiwavelength Radiative Transfer in a General Circulation Model with and without Radiatively Active Clouds. (2024). *ApJ*.

Malsky, I., Rogers, L., Kempton, E. M. R., et al. Helium-enhanced planets along the upper edge of the radius valley. (2023). *Nature Astronomy*.

Malsky, I., Rauscher, E., Kempton, E. M. R., et al. Modeling the High-resolution Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters. (2021). *ApJ*.

Malsky, I., Rogers, L. A. Coupled Thermal and Compositional Evolution of Photoevaporating Planet Envelopes. (2020). *ApJ*.

Contributing Author

Frazier, R. C., Rauscher, E., et al. (incl. **Malsky, I.**). The Days Drag On on WASP-121 b: Interpreting Its NIRISS Spectroscopic Phase Curve with General Circulation Models. (2026). *ApJ*.

Mang, J., Batalha, N. E., et al. (incl. **Malsky, I.**). PICASO 4.0: Clouds and Photochemistry in Climate Models of Brown Dwarfs and Exoplanets. (2026). *ApJ*.

Gardner, T., Monnier, J. D., et al. (incl. **Malsky, I.**). Surprise Nondetection of Upsilon Andromedae b with MIRC-X and MYSTIC at the CHARA Array. (2026). *AJ*.

van Sluijs, L., Rauscher, E., et al. (incl. **Malsky, I.**). First Results from WINERED: Detection of Emission Lines from Neutral Iron and a Combined Set of Trace Species on the Dayside of WASP-189 b. (2025). *AJ*.

van Sluijs, L., Beltz, H., **Malsky, I.**, et al. Assessing Robustness and Bias in 1D Retrievals of 3D Global Circulation Models at High Spectral Resolution: A WASP-76 b Simulation Case Study in Emission. (2025). *ApJ*.

Triantafillides, A., Savel, A. B., et al. (incl. **Malsky, I.**). Out on a Limb: The Signatures of East-West Asymmetries in Transmission Spectra from General Circulation Models. (2025). *ApJ*.

Steinrueck, M. E., Parmentier, V., et al. (incl. **Malsky, I.**). The Radiative Effects of Photochemical Hazes on the Atmospheric Circulation and Phase Curves of Sub-Neptunes. (2025). *ApJ*.

Davenport, B., Kennedy, T., et al. (incl. **Malsky, I.**). An Analysis of Spitzer Phase Curves for WASP-121b and WASP-77Ab. (2025). *AJ*.

Kennedy, T. D., Rauscher, E., **Malsky, I.**, et al. Radiatively Active Clouds and Magnetic Effects Explored in a Grid of Hot Jupiter GCMs. (2025). *ApJ*.

Smith, P. C. B., Sanchez, J. A., et al. (incl. **Malsky, I.**). The Roasting Marshmallows Program with IGRINS on Gemini South. II. WASP-121 b has Superstellar C/O and Refractory-to-volatile Ratios. (2024). *AJ*.

Monnier, J. D., Gardner, T., et al. (incl. **Malsky, I.**). Precision interferometry with MIRC-X/MYSTIC for exoplanets. (2024). *Optical and Infrared Interferometry and Imaging IX*.

Hammond, M., Bell, T. J., et al. (incl. **Malsky, I.**). Two-dimensional Eclipse Mapping of the Hot-Jupiter WASP-43b with JWST MIRI/LRS. (2024). *AJ*.

Bell, T. J., Crouzet, N., et al. (incl. **Malsky, I.**). Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b. (2024). *Nature Astronomy*.

King, G. W., Corrales, L. a. R., et al. (incl. **Malsky, I.**). The XUV-driven escape of the planets around TOI-431 and nu2 Lupi. (2024). *MNRAS*.

Kempton, E. M. R., Zhang, M., et al. (incl. **Malsky, I.**). A reflective, metal-rich atmosphere for GJ 1214b from its JWST phase curve. (2023). *Nature*.

Gao, P., Piette, A. A. A., et al. (incl. **Malsky, I.**). The Hazy and Metal-rich Atmosphere of GJ 1214 b Constrained by Near- and Mid-infrared Transmission Spectroscopy. (2023). *ApJ*.

Beltz, H., Rauscher, E., Kempton, E. M. R., **Malsky, I.**, et al. Magnetic Effects and 3D Structure in Theoretical High-resolution Transmission Spectra of Ultrahot Jupiters: the Case of WASP-76b. (2023). *AJ*.

Murphy, M. M., Beatty, T. G., Roman, M. T., **Malsky, I.**, et al. A Lack of Variability between Repeated Spitzer Phase Curves of WASP-43b. (2023). *AJ*.

Savel, A. B., Kempton, E. M. R., et al. (incl. **Malsky, I.**). Diagnosing Limb Asymmetries in Hot and Ultrahot Jupiters with High-resolution Transmission Spectroscopy. (2023). *ApJ*.

Christie, D. A., Lee, E. K. H., et al. (incl. **Malsky, I.**). CAMEMBERT: A Mini-Neptunes General Circulation Model Intercomparison, Protocol Version 1.0.A CUISINES Model Intercomparison Project. (2022). *PSJ*.

Beltz, H., Rauscher, E., Kempton, E. M. R., **Malsky, I.**, et al. Magnetic Drag and 3D Effects in Theoretical High-resolution Emission Spectra of Ultrahot Jupiters: the Case of WASP-76b. (2022). *AJ*.

Gardner, T., Monnier, J. D., et al. (incl. **Malsky, I.**). A thesis to probe unique exoplanet regimes with micro-arcsecond astrometry and precision closure phases at CHARA and VLTI. (2022). *Optical and Infrared Interferometry and Imaging VIII*.

Caldiroli, A., Haardt, F., et al. (incl. **Malsky, I.**). Irradiation-driven escape of primordial planetary atmospheres. II. Evaporation efficiency of sub-Neptunes through hot Jupiters. (2022). *A&A*.

Belkovski, M., Becker, J., Howe, A., **Malsky, I.**, et al. A Multiplanet System's Sole Super-puff: Exploring Allowable Physical Parameters for the Cold Super-puff HIP 41378 f. (2022). *AJ*.

Caldiroli, A., Haardt, F., et al. (incl. **Malsky, I.**). Irradiation-driven escape of primordial planetary atmospheres. I. The ATES photoionization hydrodynamics code. (2021). *A&A*.

Harada, C. K., Kempton, E. M. R., et al. (incl. **Malsky, I.**). Signatures of Clouds in Hot Jupiter Atmospheres: Modeled High-resolution Emission Spectra from 3D General Circulation Models. (2021). *ApJ*.

Kasper, D., Bean, J. L., Oklopčić, A., **Malsky, I.**, et al. Nondetection of Helium in the Upper Atmospheres of Three Sub-Neptune Exoplanets. (2020). *AJ*.

Software

Batalha, N., Mang, J., et al. (incl. **Malsky, I.**). PICASO Release 4.0. (2026). *Zenodo*.

Savel, A., M.-R. Kempton, E., Beltz, H., **Malsky, I.** arjunsavel/hires-literature. (2024). *Zenodo*.

Malsky, I., Cowherd, M., Rauscher, E., et al. emily-rauscher/RM-GCM: First public version of RM-GCM. (2023). *Zenodo*.

vibe-check (github.com/imalsky/vibe-check): reliability checks for machine-learning surrogates in scientific software; supported by URSSI.

VULCAN-JAX (github.com/imalsky/jax-vulcan): JAX port of the VULCAN chemical-kinetics code, working toward a fully auto-differentiable version.

JWST Transit Authority (github.com/imalsky/jwst-transit-authority): JWST observation planner that combines VULCAN-JAX chemistry, ExoJAX spectra, and Pandeia noise to rank instrument modes.

Problemulator (github.com/imalsky/Problemulator): transformer emulator for exoplanet radiative transfer; code for Malsky et al. (in review).

Chemulator (github.com/imalsky/Chemulator): neural-network emulator for stiff chemical kinetics; code for Malsky et al. (2026, *ApJ*).

Grants and Awards

2026 URSSI Early-Career Fellowship, U.S. Research Software Sustainability Institute

2024–2026 NASA Postdoctoral Program Fellowship

2023–2024 Rackham Predoctoral Fellowship

2023 Rackham Professional Development Grant

2023 Rackham Graduate Student Research Grant

2021 NSF Graduate Research Fellowship Program Honorable Mention

2021 Michigan Space Grant

2015 Jeff Metcalf Fellowship

2014 National Merit Scholarship

2014–2018 University of Chicago Dean's List

Mentoring

2025–2026 Yashnil Mohanty, high school project

2024–2025 Elijah Mullens, graduate research

2023–2025 Thomas Kennedy, graduate research

2022–2024 Alex Wingate, undergraduate research

2022–2024 Eli Cinque, undergraduate research

Teaching

Spring 2021 ASTRO 105: Cosmos Constellations, Graduate Student Instructor, University of Michigan

Fall 2020 ASTRO 101: Introductory Astronomy: The Solar System and the Search for a New Earth, Graduate Student Instructor, University of Michigan

Service, Outreach, and Skills

Peer review *The Astrophysical Journal*; *The Astrophysical Journal Letters*; *The Planetary Science Journal*; *The Journal of Open Source Software*.

Service Intro2Astro Course Volunteer Summer (2021)
Scarlett Middle School Teaching Assistant (2020)
Science Olympiad Mentor at Slauson Middle School (2020)
Ann Arbor Astronomy on Tap (2019)
Invisible Institute Volunteer Infographic Designer (2018).

Training Exoclimes Summer School (2023)
SC23 Supercomputing Workshops (2023)
University of Chicago Supercomputer Workshops (2018).

Computing Python (PyTorch, JAX, NumPy, SciPy), Fortran, git, GPU training, HPC systems (Slurm/PBS). Basic: C.

Observing Proposals

JWST **[Co-I]** Cloudy with a Chance of Disequilibrium Chemistry: A Comprehensive View of Silicate Clouds in a Hot Jupiter with a MIRI/LRS Phase Curve, Splinter et al., GO 11301, Cycle 5, 69.6 hours.

JWST **[Co-I]** Meeting their Match: Measuring Cloudy Hot Jupiter Counterparts to Understand the Silicate Cloud Regime with JWST/MIRI, Moran et al., GO 7849, Cycle 4, 49.1 hours.

Magellan **[Co-I]** The Beginning of a High-Resolution Emission Spectroscopy Program at Magellan for the Characterization of Hot Jupiter Atmospheres with the WINERED Instrument, Sluijs et al., 3 nights (2023).

Gemini South **[Co-I]** Tracing the Day-Night Structure of WASP-121b with Multi-Phase High-Resolution Spectroscopy, Rauscher et al., 12 hours (2023).

Gemini South **[Co-I]** Tracing the Day-Night Structure of WASP-76b with Multi-Phase High-Resolution Spectroscopy, Rauscher et al., 20 hours (2023).

Magellan **[Co-I]** The First High-Resolution Emission Spectrum of an Exoplanet with WINERED at Magellan, Rauscher et al., 12 hours (2022).

Selected Presentations

2026 **ERES.** AI for Exoplanet Modeling/Analysis. (talk)

2026 **Yuk Yung Lunch Talk.** AI for Exoplanet Modeling/Analysis. (invited talk)

2026 **Montana State University Colloquium.** AI for Exoplanet Modeling/Analysis. (invited talk)

2026 **MIT Monday Afternoon Talks.** AI for Exoplanet Modeling/Analysis. (invited talk)

2026 **ExoPAG 33.** AI for Exoplanet Modeling/Analysis. (invited talk)

2025 **ExSoCal.** Breaking the Accuracy-Speed Trade-off in Radiative Transfer. (talk)

2025 **Aspen Rocky Worlds.** Clouds and Hazes in GJ 1214 b's Metal-Rich Atmosphere. (talk)

2025 **NASA Ames.** ML Emulation of Atmospheric Radiative Transfer. (talk)

2024 **Cornell University.** Clouds and Hazes in GJ 1214 b's Metal-Rich Atmosphere. (talk)

2024 **NASA Jet Propulsion Laboratory.** Characterizing Exoplanet Atmospheres: 1D and 3D. (talk)

2024 **Los Alamos National Laboratory.** Characterizing Exoplanet Atmospheres: 1D and 3D. (talk)

2023 **Exoclimes VI.** Clouds and Hazes in Polychromatic GCMs. (talk)
2022 **Bay Area Exoplanets Meeting.** Clouds and Hazes in Polychromatic GCMs. (talk)
2022 **The Other Worlds Laboratory.** Double Gray vs. Multiwavelength GCMs. (talk)
2021 **GLEAM.** Modeling the 3D Atmospheric Structure of Hot Gaseous Planets. (talk)
2020 **Exoplanets IV.** Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters. (poster)
2018 **MIT-Harvard.** The Coupled Thermal Evolution of Sub-Neptune Atmospheres. (poster)
2018 **US Naval Academy.** The Coupled Thermal Evolution of Sub-Neptune Atmospheres. (poster)